# Aprendizagem Aplicada à Segurança

## Class 5: Anomaly Detection I, statistics, neighbours, clusters and classifiers

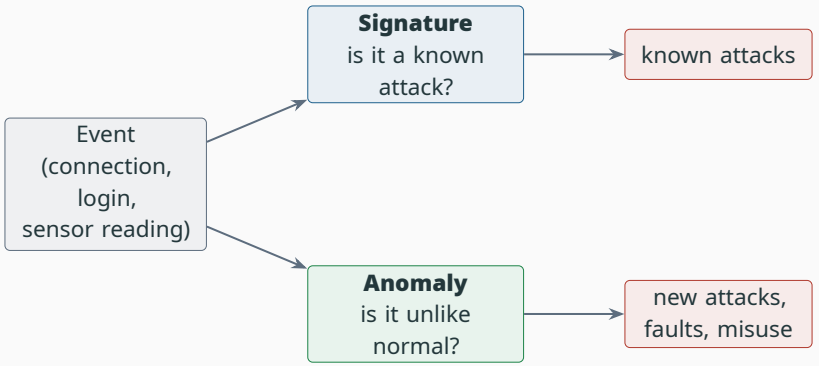
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# Signatures know the past, anomalies look for the rest

- An **event** is one observation that the detector judges: a network connection, a login, a sensor reading.
- A **signature** (misuse) detector matches known attacks: precise, but blind to anything new.
- An **anomaly** detector models *normal* behaviour and alarms on departures from it.



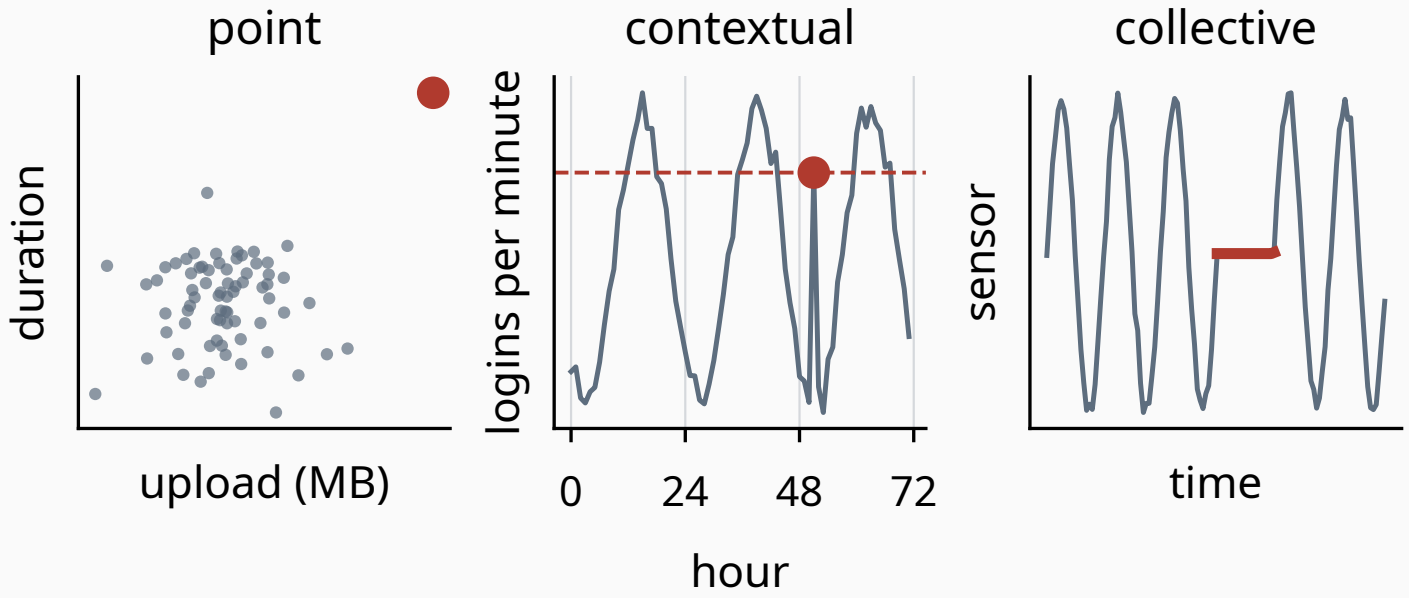

- The price: rare but legitimate events are false alarms.

# Terminology: the words of this class

| Term | Meaning | Symbol | Example |
|---|---|---|---|
| normal | behaviour seen in the training data | $X$ | ordinary sessions of one host |
| anomaly | an event unlike normal; rare, not always malicious | | a 2 GB upload |
| score | number, higher = more unusual | $s(\mathbf{x})$ | distance from normal |
| threshold | alarm when the score exceeds it | $t$ | $t = 9.21$ |
| TPR, FPR | share of attacks, of normal events, that raise an alarm | TPR, FPR | 0.95, 0.01 |
| base rate | share of attacks among all events | $b$ | 1 in 10,000 |

# Three kinds of anomaly

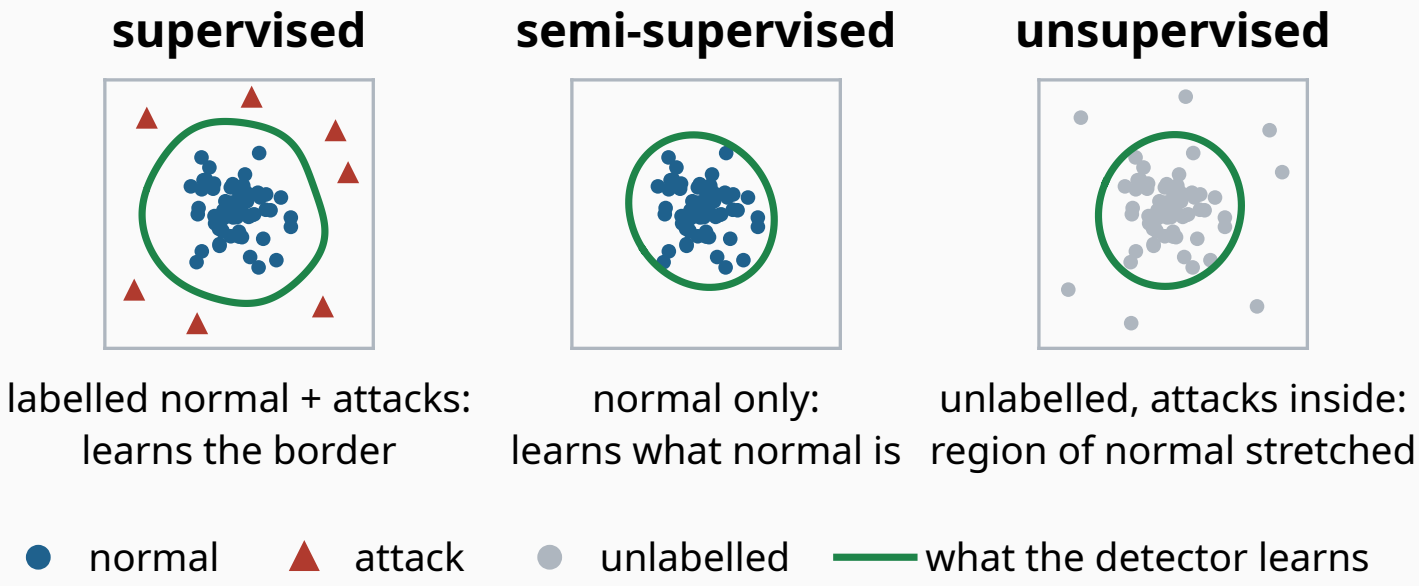
Schematic: unusual alone, in its context, or as a group.
Today: point anomalies (Chandola et al. 2009).




- **Point:** one event far from the rest (a 2 GB upload).
- **Contextual:** a normal value at the wrong time.
- **Collective:** each value ordinary, the pattern is not.

# Three settings, by what the training data holds

Schematic: the same events, other labels (Sommer and Paxson 2010). Which labels do we have?




- **Supervised:** labels for normal and attacks.
- **Semi-supervised (novelty):** only normal data.
- **Unsupervised:** no labels; attacks inside.

# Families of detectors: the map of this class

| Family | Normal is… | Methods |
|---|---|---|
| distribution-based | a probability distribution | z-score, Mahalanobis distance, Gaussian mixture |
| distance and density | a dense region | kNN distance, LOF |
| isolation | hard to separate by cuts | Isolation Forest |
| boundary | inside a learned frontier | one-class SVM |
| reconstruction | close to a subspace or to a few parts | PCA, ICA, NMF (autoencoders: class 03_02) |
| clusters | close to a profile | k-means, DBSCAN |
| supervised | the side of the labelled normal | logistic regression, random forest |

# Notation

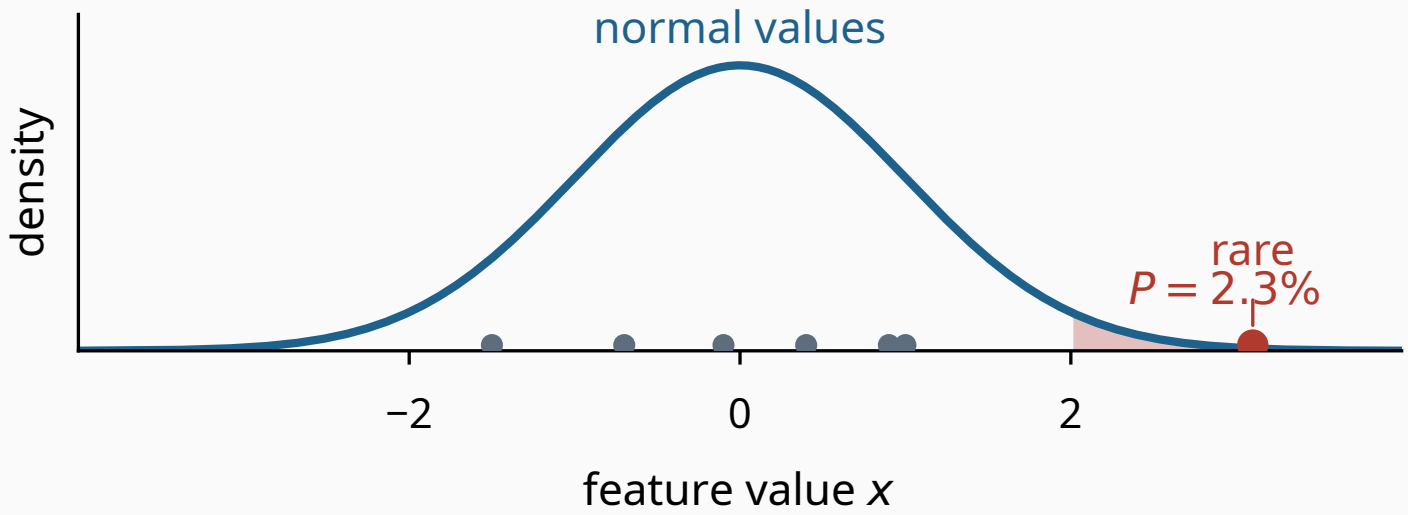
| Symbol | Meaning | Shape | Example 1 |
|---|---|---|---|
| $\mathbf{x}$ | one event as a feature vector | $d$ | $(2, 2)$ |
| $X$ | data matrix (example $i$, feature $j$): the normal training rows | $n \times d$ | 500 sessions |
| $\mu$ | mean of the normal rows | $d$ | $(0, 0)$ |
| $\sigma_j$ | standard deviation of feature $j$ | scalar | $1$ |
| $C$ | covariance matrix: how features vary together | $d \times d$ | $\begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}$ |
| $s(\mathbf{x}),\ t$ | score, threshold | scalar | $t = 9.21$ |
| $\alpha$ | significance level | scalar | $0.01$: 1% false alarms |

# Notation: distances and alarms

| Symbol | Meaning | Shape | Example 1 |
|---|---|---|---|
| $d_M$ | distance from the mean, scaled by $C^{-1}$ | scalar | $d_M(A) = 2.05$ |
| $C^{-1}$ | inverse covariance matrix | $d \times d$ | $\frac{1}{0.19}\begin{pmatrix} 1 & -0.9 \\ -0.9 & 1 \end{pmatrix}$ |
| TPR | share of attacks that raise an alarm | scalar | $0.95$ |
| FPR | share of normal events that raise an alarm | scalar | $0.01$ |
| $P(\text{attack} \mid \text{alarm})$ | share of alarms that are attacks | scalar | $0.94\%$ at $b = 10^{-4}$ |

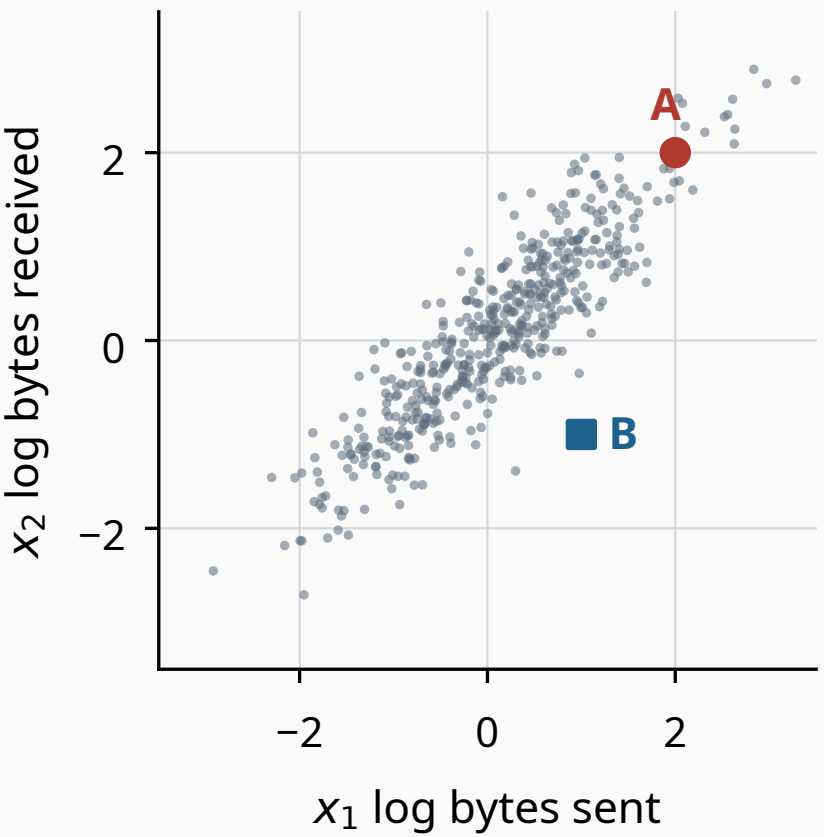
# Model normal with a distribution

- **Distribution-based detection** fits a distribution to the normal rows; the score is how unlikely an event is.
- One feature: the **z-score** $z = (x - \mu)/\sigma$, standard deviations from the mean.




- A rare value sits in the tail: $|z| > 2$ holds for 4.6% of normal events.

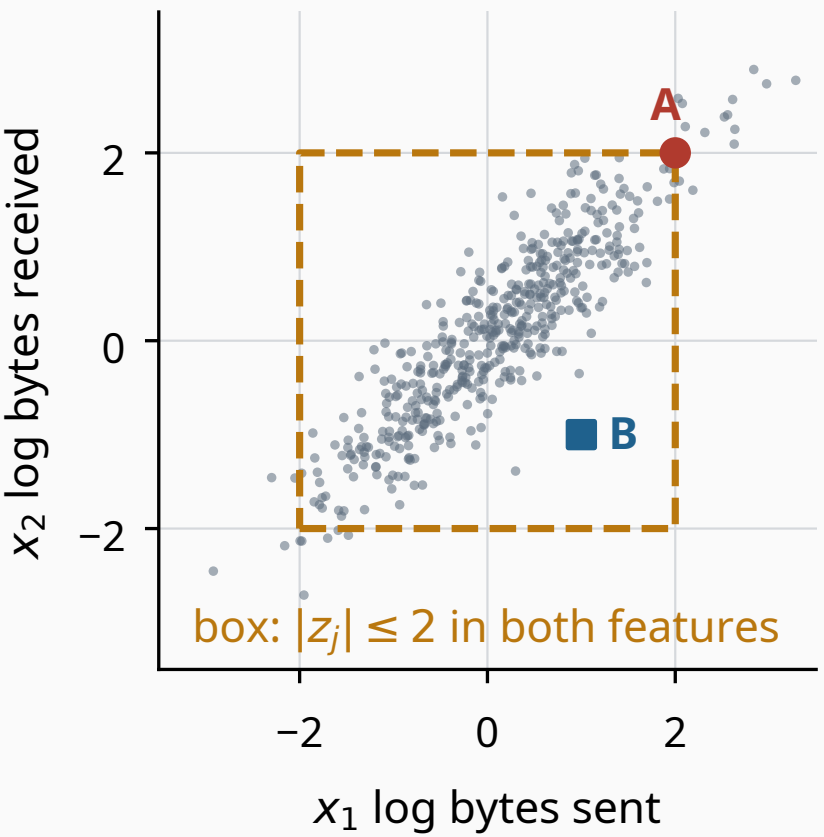
# Example 1: one host, two features




- One point = one session: log bytes sent ($x_1$) and received ($x_2$), standardised.
- Normal: $\mu = (0, 0)$, unit variances, $\rho = 0.9$.
- **A** $= (2, 2)$: big but consistent.
- **B** $= (1, -1)$: sends a lot, receives little.

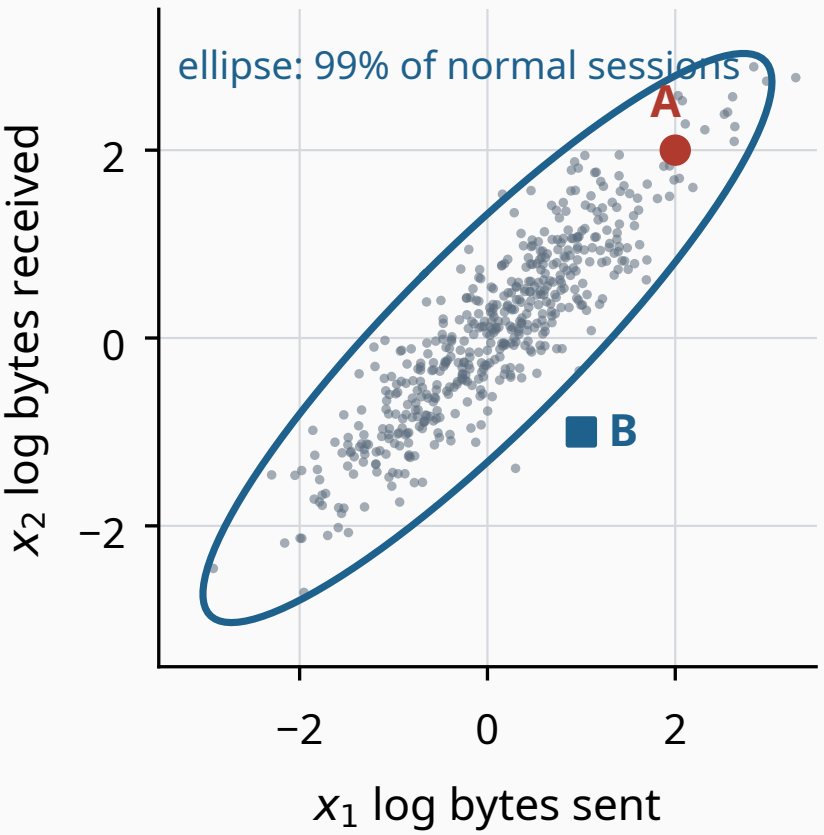
- Which is the anomaly? Notebook 00 rebuilds every number.

# The z-score looks at one feature at a time




- Score $s(\mathbf{x}) = \max_j |z_j|$: the box is $|z_j| \le 2$ in both features.
- A: $z = (2, 2)$, score **2**. B: $z = (1, -1)$, score **1**.
- The z-score ranks A above B: B hides inside the box.
- It ignores that the two features move together; B breaks the relation, not a range.

# The Mahalanobis distance: the idea



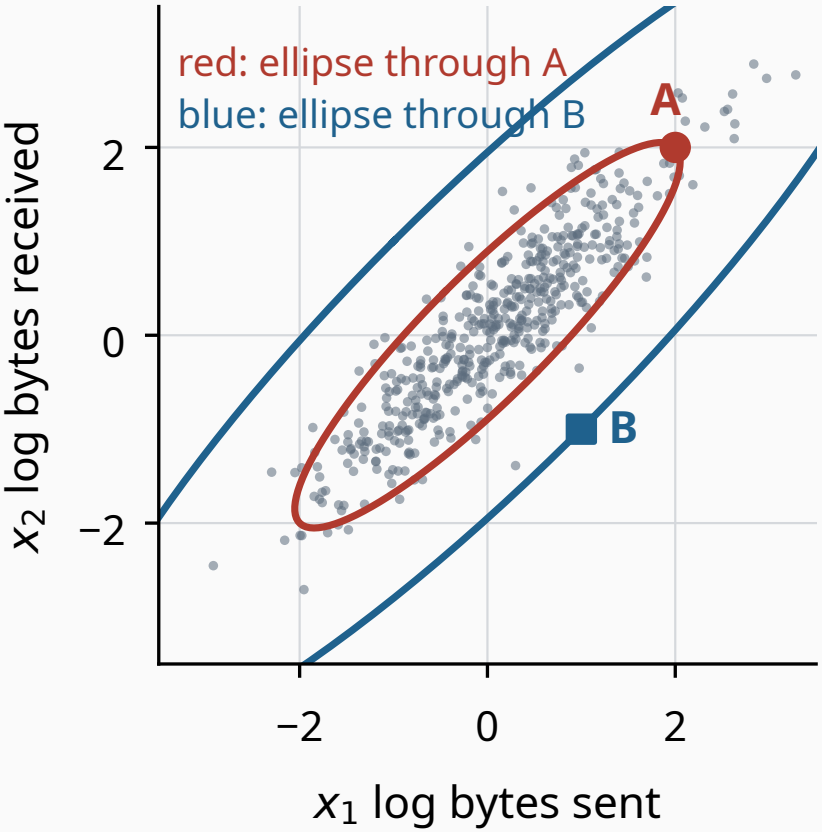

- The **Mahalanobis distance** is the distance from the mean in units of the data's own spread, along its own directions.
- Equal distance = an **ellipse** that follows the cloud, not a box.
- The ellipse $d_M = 3.03$ holds 99% of normal sessions.

# The Mahalanobis distance: definition

$$d_M(\mathbf{x})^2 = (\mathbf{x} - \mu)^\top C^{-1} (\mathbf{x} - \mu) \quad \text{(definition)}$$

- Example 1: $\det C = 1 - 0.81 = 0.19$ and $C^{-1} = \frac{1}{0.19} \begin{pmatrix} 1 & -0.9 \\ -0.9 & 1 \end{pmatrix}$.
- So $d_M^2 = (x_1^2 - 1.8\, x_1 x_2 + x_2^2)/0.19$: the cross term **rewards** features that move together.
- If $C = I$ it is the Euclidean distance; with a diagonal $C$, $d_M^2 = \sum_j z_j^2$ (the z-scores, combined).

# Example 1 by hand: A is normal, B is not

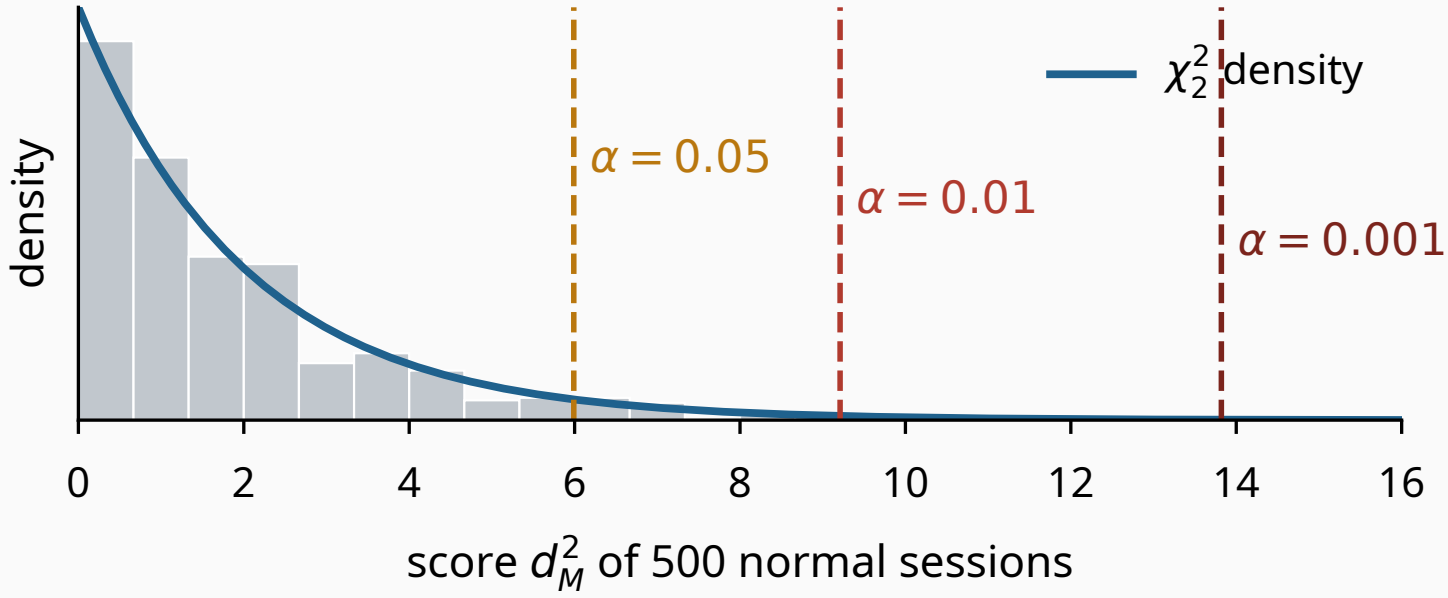



- A: $(4+4-1.8\cdot 4)/0.19 = 4.21.$
- B: $(1+1+1.8)/0.19 = 20.0.$

| Point | $d_M$ | $P(\chi^2_2 > d^2_M)$ |
|---|---|---|
| A | 2.05 | 0.12 |
| B | 4.47 | 0.000045 |

- The ranking flips: B is a 1-in-22,000 event, A a 1-in-8 event. If normal data are Gaussian, $d^2_M \sim \chi^2_d$ (appendix).

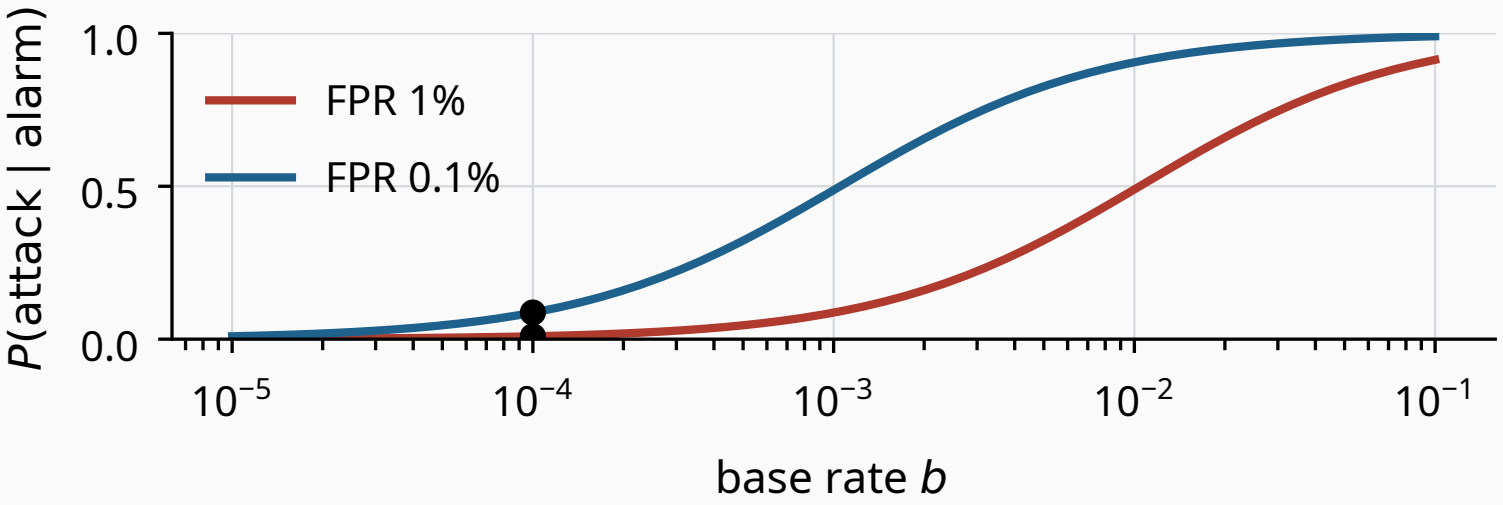
# Choosing the threshold without attacks




- Set $t$ on **normal** data: the $(1 - \alpha)$ quantile of the normal scores gives about $\alpha$ false alarms.
- $\alpha = 0.05, 0.01, 0.001$: $\chi^2_2$ gives 5.99, 9.21, 13.82; 500 sessions give 5.97, 8.86, 14.98.
- Small $\alpha$ rests on one or two extreme rows: it needs much more normal data.

# The base rate decides what an alarm is worth

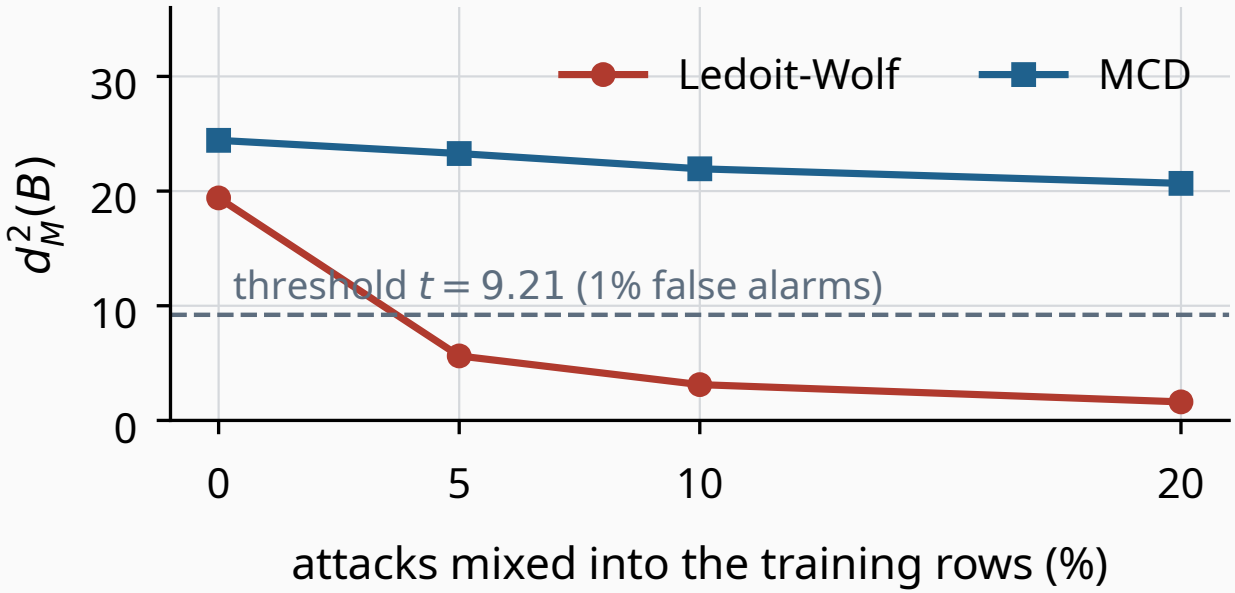
$$P(\text{attack} \mid \text{alarm}) = \frac{\text{TPR} \cdot b}{\text{TPR} \cdot b + \text{FPR} \cdot (1 - b)} \quad \text{(Bayes' rule)}$$




- TPR 0.95, $b = 10^{-4}$, FPR 1%: **99 of 100 alarms are false** (Axelsson 2000).

# Contamination and robust estimates

- **Contamination:** the training rows already hold attacks (copies of B); the mean and covariance follow them.

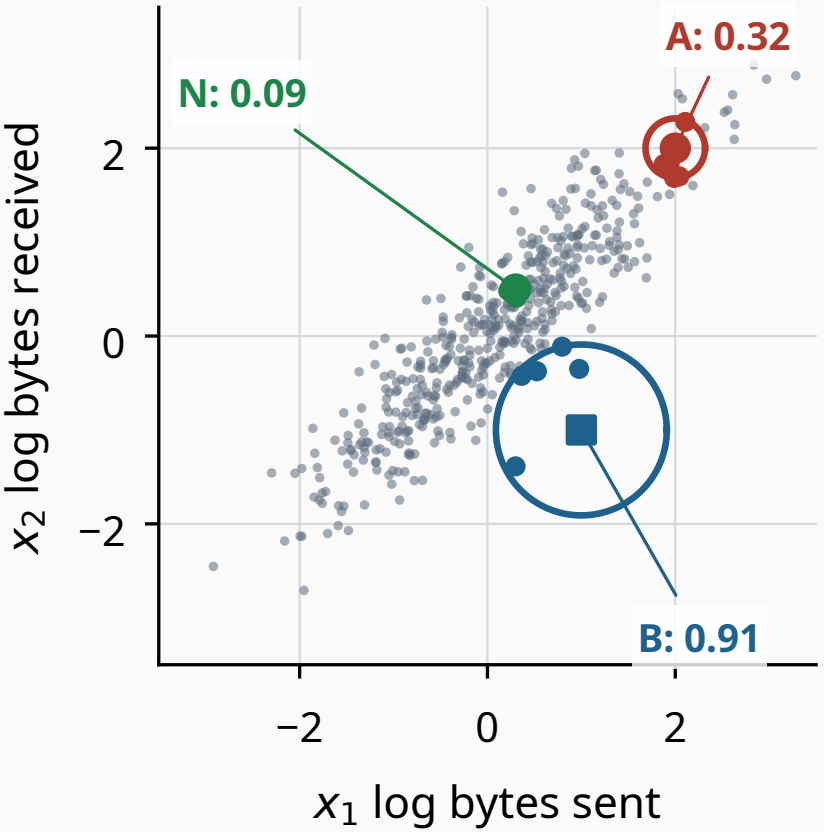



- **Ledoit-Wolf** (Ledoit and Wolf 2004) is not robust: 5% copies cut $d_M^2(B)$ from 19.4 to 5.6; **MCD** (Rousseeuw and Van Driessen 1999): 23.3.

# Notation: neighbours, trees and subspaces

| Symbol | Meaning | Shape | Used in |
|---|---|---|---|
| $k,\ N_k(\mathbf{x})$ | number of neighbours, the $k$ nearest neighbours of **x** | integer, set | kNN, density |
| **o** | one neighbour of **x** | $d$ | density |
| $\mathrm{LOF}_k(\mathbf{x})$ | neighbours' density over its own | scalar | density |
| $h$ | path length in an isolation tree | scalar | forest |
| $c(n)$ | mean path length in a random tree of $n$ points | scalar | forest |
| $V_k$ | the top $k$ principal directions | $d \times k$ | PCA |

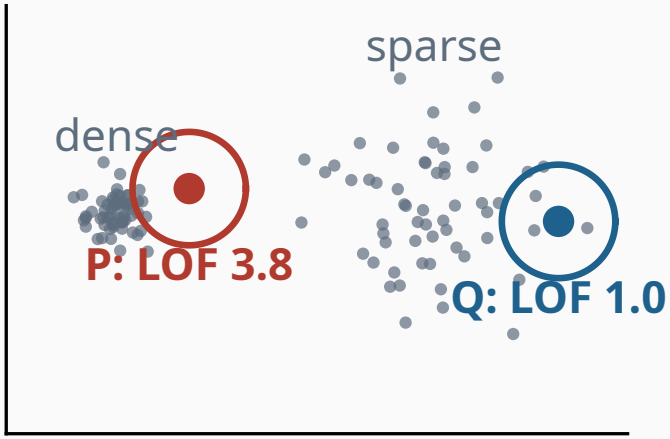
# Distance: the kNN score




- $s(\mathbf{x})$ = distance to the $k$-th nearest normal point ($k = 5$): the radius of each circle.
- N (normal): 0.09. A: 0.32.
- B: 0.91, the largest circle: B sits in a sparse part of the cloud.

# Density: the idea of LOF

- **Local Outlier Factor** (LOF) (Breunig et al. 2000): density around **x** against density around its neighbours.



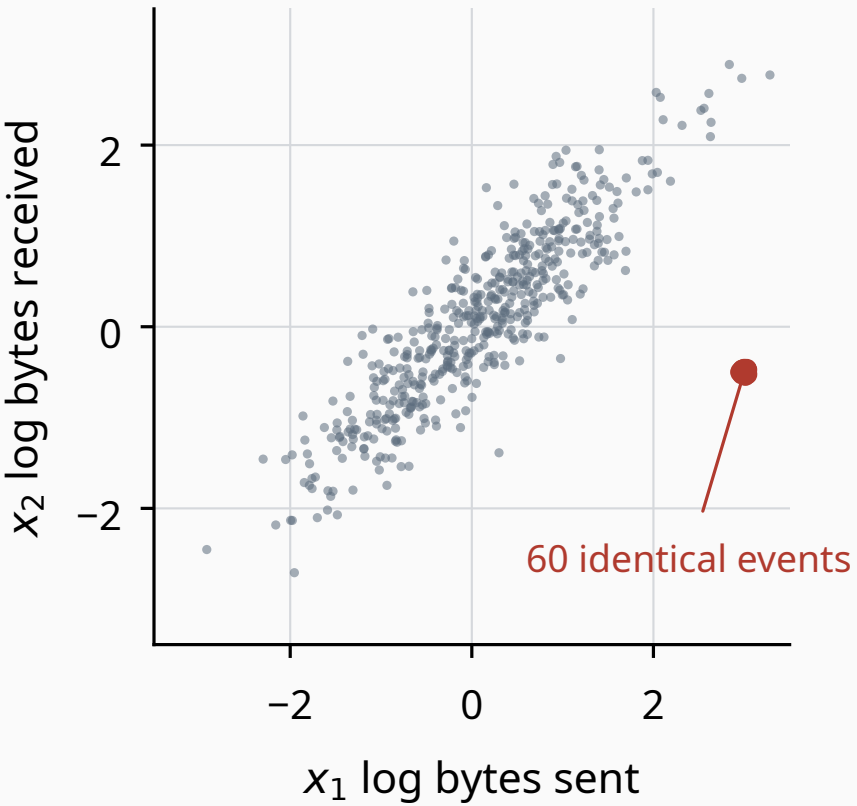

- P and Q have the same kNN distance (0.82).
- P is far from its dense neighbours: LOF 3.8. Q: 1.05 ($\approx 1$ is normal).

# LOF: definition

$$\mathrm{LOF}_k(\mathbf{x}) = \frac{1}{k} \sum_{\mathbf{o} \in N_k(\mathbf{x})} \frac{\mathrm{lrd}(\mathbf{o})}{\mathrm{lrd}(\mathbf{x})} \quad \text{(definition)}$$

- $N_k(\mathbf{x})$: the $k$ nearest neighbours. lrd: local reachability density, the inverse of the mean distance to them (appendix).
- A ratio of densities: it adapts to regions of different density, and returns 1 inside any dense cluster.

# LOF is blind to a dense attack

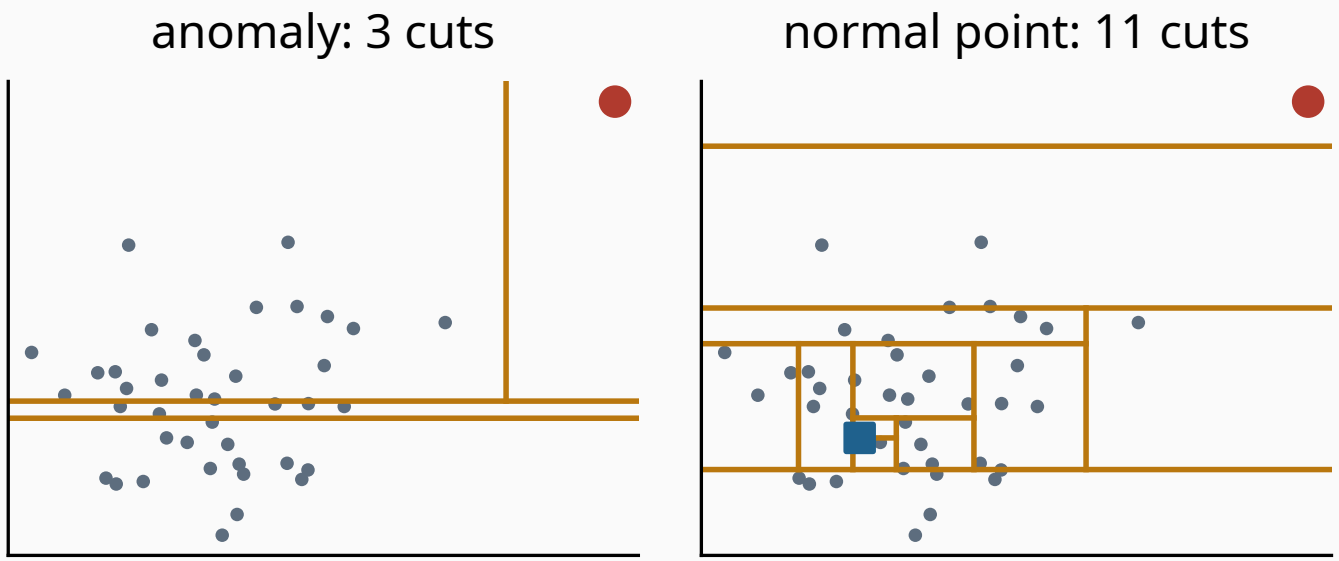



- A flood is many near-identical events: a **dense** cluster (red, 60 points).
- LOF of the flood: 5.66 at test time, **1.07** in training: its points are each other's neighbours.
- On KDD99, LOF misses 92% of the DoS rows.

Notebook 00, part D rebuilds these numbers.

# Isolation: the idea

- **Isolation Forest** (Liu et al. 2008) cuts a random feature at a random value until the event is alone.
- An anomaly is alone after **few** cuts; an event in a dense region needs many.



# Isolation Forest: definition and score

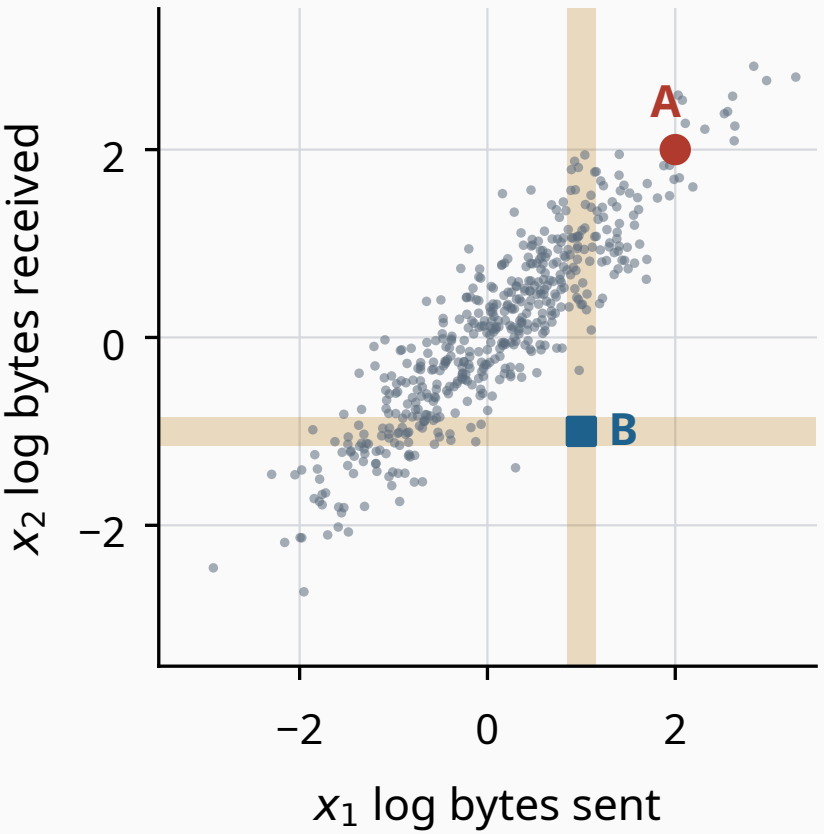
$$s(\mathbf{x}) = 2^{-E[h(\mathbf{x})]/c(n)} \qquad \text{(definition)}$$

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n} \qquad \text{(definition)}$$

- $h$: path length (number of cuts); $E[h]$: its mean over the trees.
- $c(n)$: mean path in a random binary tree of $n$ points; $H(i) \approx \ln i + 0.5772$.
- Trees use subsamples of $n = 256$: $c(256) = 10.25$. Isolated after 4 cuts: $s = 2^{-4/10.25} = 0.76$; after 10.25 cuts: $s = 0.5$.

# Isolation Forest: fast, robust, axis-bound

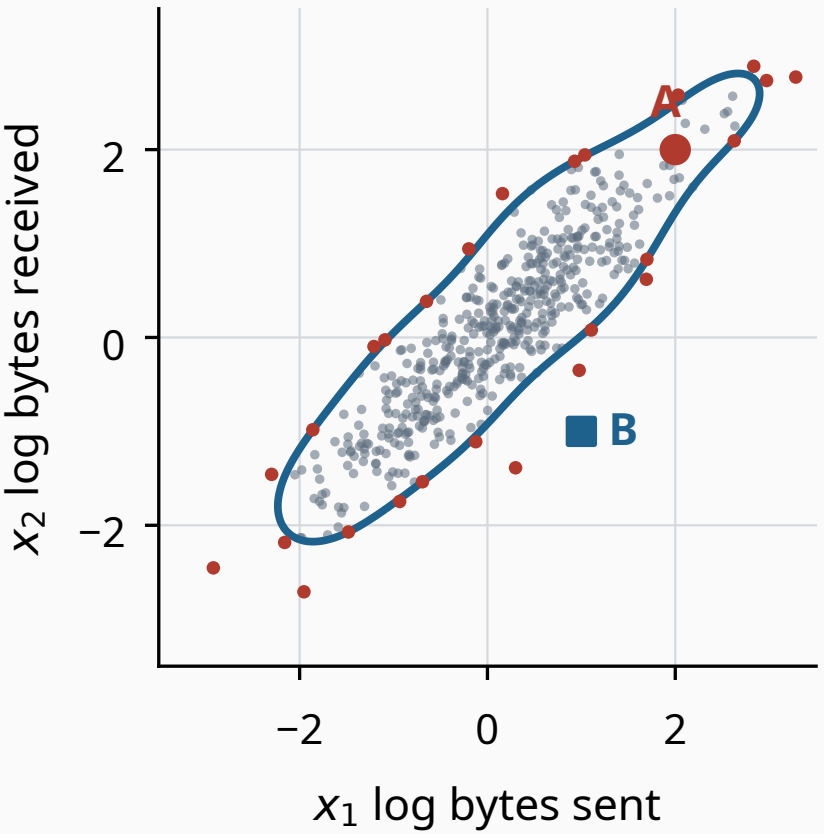
- Linear time; each tree sees 256 random rows: robust to contamination.




- Cuts are **axis-aligned**.
- B is ordinary on each axis (41 and 39 normal sessions share its $x_1$, $x_2$): no cut isolates it.
- Rank of B among 500 normal sessions (0 = above all): Mahalanobis 0, LOF 0, kNN 2, **forest 31**.

# A boundary around normal: the one-class SVM

- The **one-class SVM** (Schölkopf et al. 2001) learns a frontier that encloses most of the normal rows.




- RBF kernel, $\nu = 0.05$: 27 of 500 rows (red) fall outside.
- B is far outside, A inside.
- $\nu$ bounds the share of rows left outside: the target false-alarm rate.

# One-class SVM: definition

- Map the data with a kernel map $\phi$; separate it from the **origin** with the widest margin:

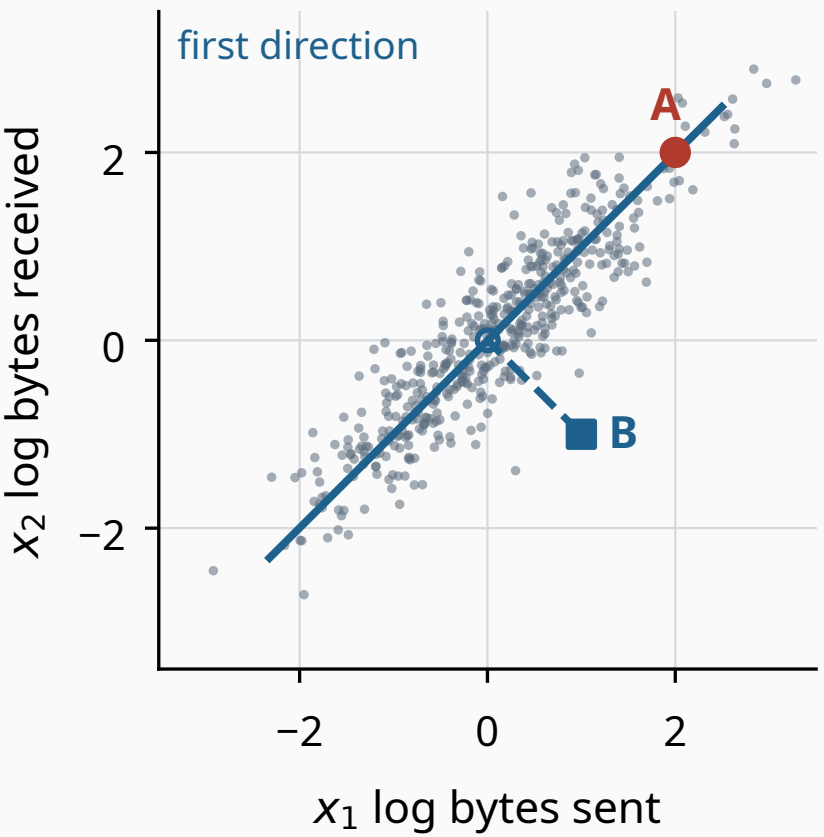
$$\min_{\mathbf{w},\xi,\rho} \ \tfrac{1}{2}\|\mathbf{w}\|^2 + \frac{1}{\nu n}\sum_i \xi_i - \rho$$

$$\text{subject to} \quad \mathbf{w}^\top \phi(\mathbf{x}_i) \geq \rho - \xi_i, \quad \xi_i \geq 0$$

- $\xi_i$: slack of row $i$, how far it falls outside. Score: $s(\mathbf{x}) = \rho - \mathbf{w}^\top \phi(\mathbf{x})$, positive outside.
- Training is quadratic in $n$: we fit 5,000 rows.

# A subspace for normal: PCA reconstruction

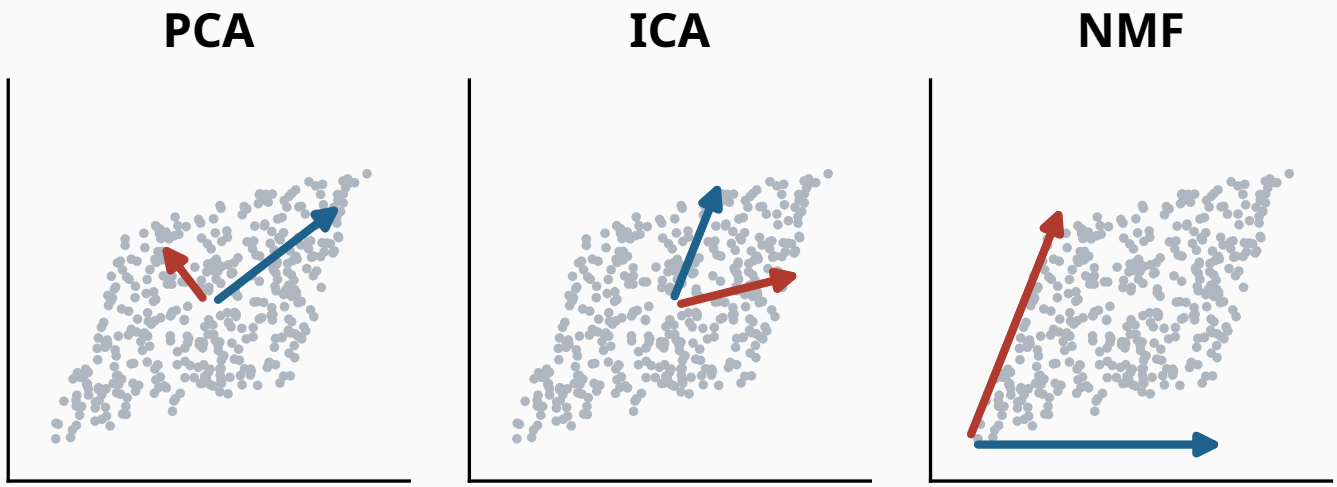
$$s(\mathbf{x}) = \|(\mathbf{x} - \mu) - V_k V_k^\top (\mathbf{x} - \mu)\|^2 \quad \text{(definition)}$$




- $V_k$ ($d \times k$): top $k$ principal directions ($k = 1$, blue line).
- Score = the dashed residual: A 0.0, B 1.99.
- PCA is the linear twin of the autoencoder (class 03_02).

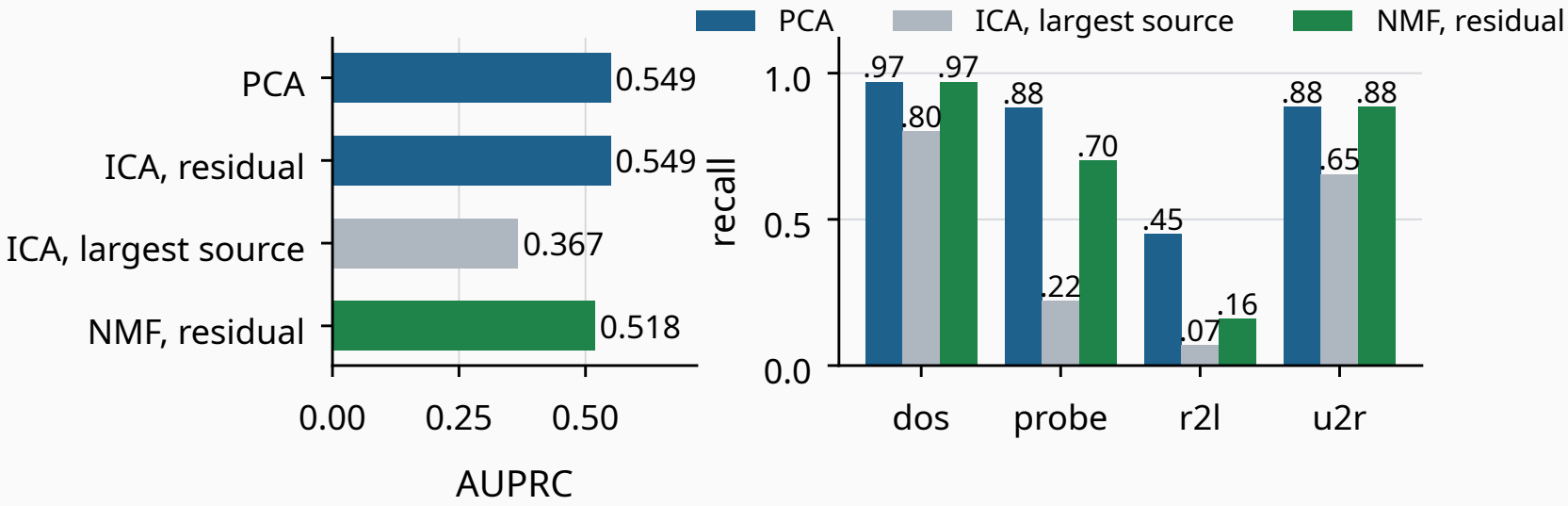
# Blind source separation: PCA, ICA and NMF

- **ICA** (independent component analysis (Hyvärinen and Oja 2000)) and **NMF** (non-negative matrix factorisation (Lee and Seung 1999)) join PCA.
- Each event is a mix of a few sources or parts.
- The constraint differs; the score is always the squared residual.



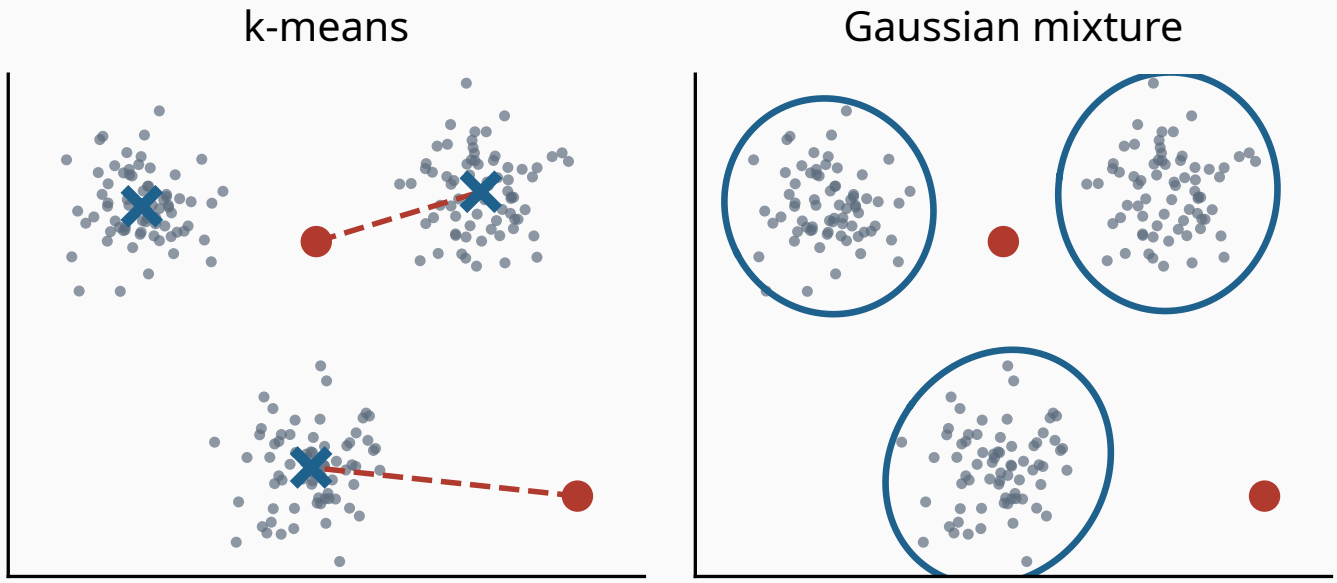
# PCA, ICA and NMF on KDD99

Question: does the constraint change what is found?



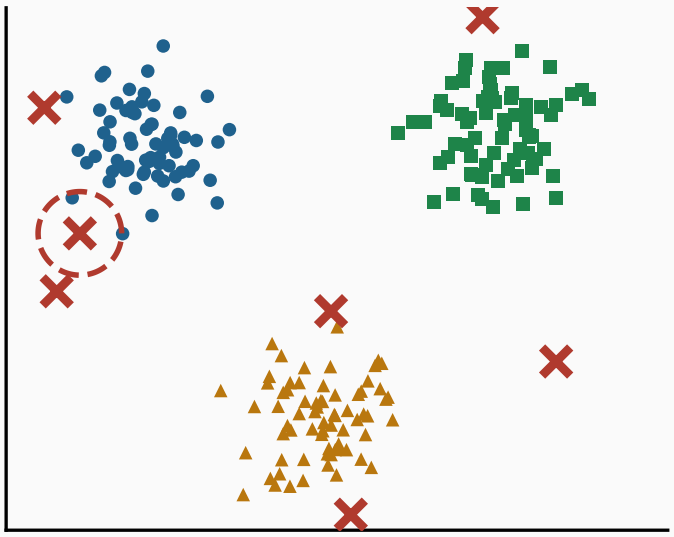

- $k = 41$ (95% of the variance); NMF by `pyNNMF` (HALS). ICA spans PCA's subspace: residual 0.549, largest source 0.367.
- NMF (0.518) misses probe (0.70) and r2l (0.16).

# Clusters: k-means and mixtures




- Three profiles (web, mail, DNS). **k-means**: distance to the nearest centre (far point 4.0, between profiles 2.57).
- **Gaussian mixture**: $s = -\log p(\mathbf{x})$, $p = \sum_j \pi_j \mathcal{N}(\mathbf{x}; \mu_j, C_j)$: one Mahalanobis model per profile. Red points: 28.8 and 12.7 (normal: 2.4).

# Clusters: DBSCAN



- **DBSCAN** (Ester et al. 1996) marks an event as *noise* (red) when fewer than 6 events lie within $\varepsilon = 0.6$ (dashed circle).
- An anomaly label for free, but no score for a new event.
- Normal traffic is a mixture (web, mail, DNS): clusters fit it better than one Gaussian.

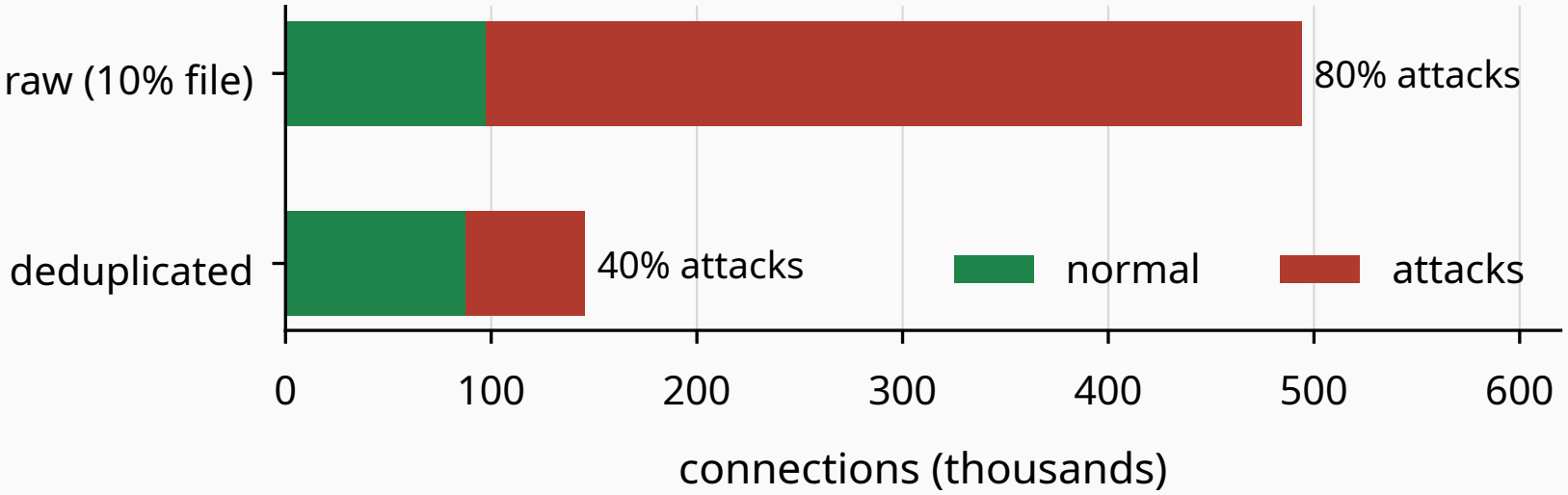
# A famous, flawed benchmark

- 41 features per TCP/IP connection from simulated 1998 traffic; 22 attacks in four categories.

| Category | What it does | Examples |
|---|---|---|
| dos | floods a service | neptune (SYN flood), smurf |
| probe | scans hosts and ports | satan, ipsweep, nmap |
| r2l | gets a local account from outside | guess_passwd, warezclient |
| u2r | escalates to root | buffer_overflow, rootkit |

- Known flaws: duplicates, too easy, old traffic (Tavallaee et al. 2009). We use it *because* its flaws teach the protocol.

# Lesson 1: remove the duplicates




- 70.5% of the rows are copies, mostly of two floods (smurf, neptune): a detector is graded on spotting them.
- Deduplicated: 87,831 normal, 54,572 dos, 2,131 probe, 999 r2l, 52 u2r.
- Even so, 40% attacks is far from any real network.

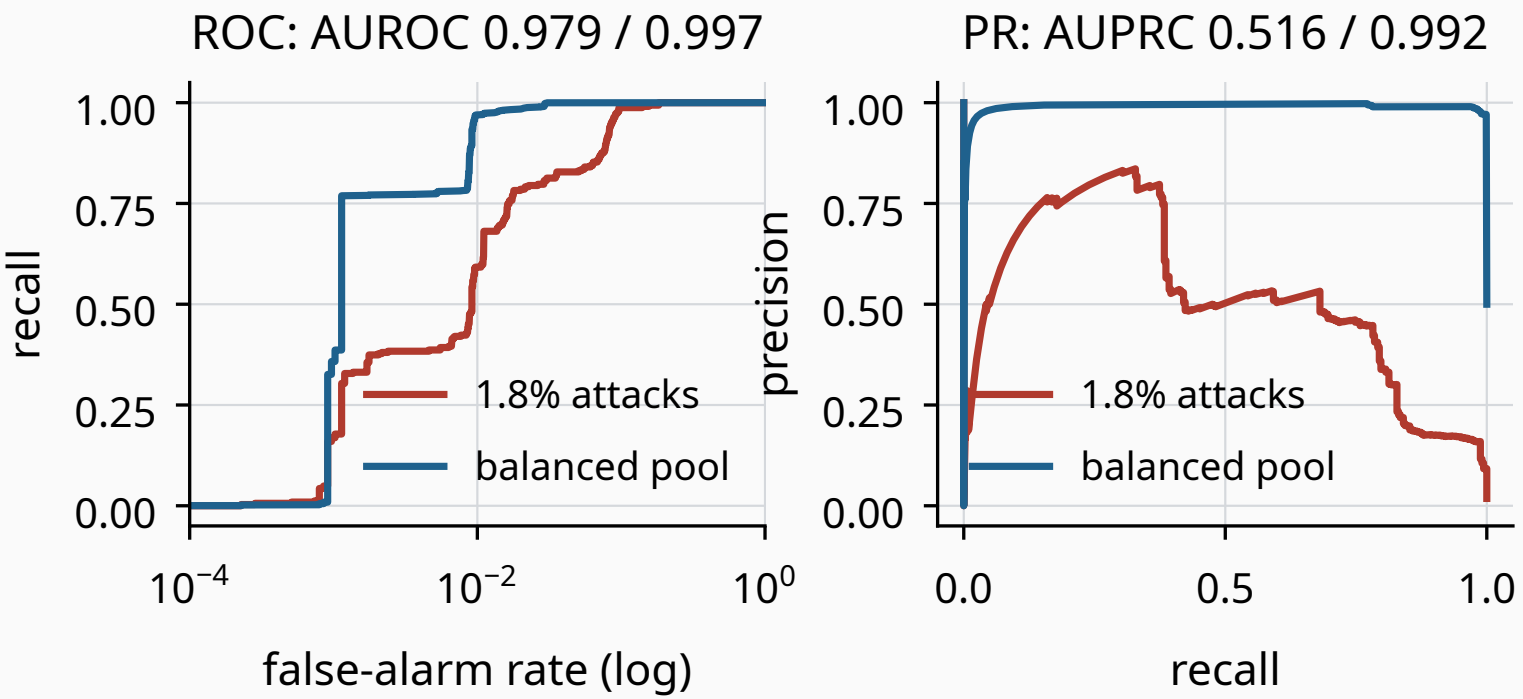
# Terminology: the metrics

| Metric | Question it answers | Mahalanobis, test set |
|---|---|---|
| recall (TPR) | which share of the attacks raises an alarm? | 0.681 |
| false-alarm rate (FPR) | which share of normal events raises an alarm? | 0.010 |
| **precision** | which share of the alarms is an attack? | 0.516 |
| **AUROC** | does a random attack outscore a random normal event? | 0.979 |
| **AUPRC** | precision averaged over all recalls; a random score gets $b$ | 0.516 |
| recall per category | recall inside dos, probe, r2l, u2r | next frames |

# The protocol: novelty detection at a realistic base rate

- Features: $\log(1+x)$ for bytes and counts, one-hot for protocol, service, flag (71 features), z-scaling fitted on normal training rows.
- Normal rows 60 / 20 / 20: **train** (52,698), **validation** (17,566), **test**; attacks never in training.
- Test attacks: 100 per category (u2r: 26), **1.8%** of the test set; the rest form a labelled **pool**.
- Threshold: 99% quantile of the validation scores (1% false alarms).

# Lesson 2: ROC hides the base rate, precision does not



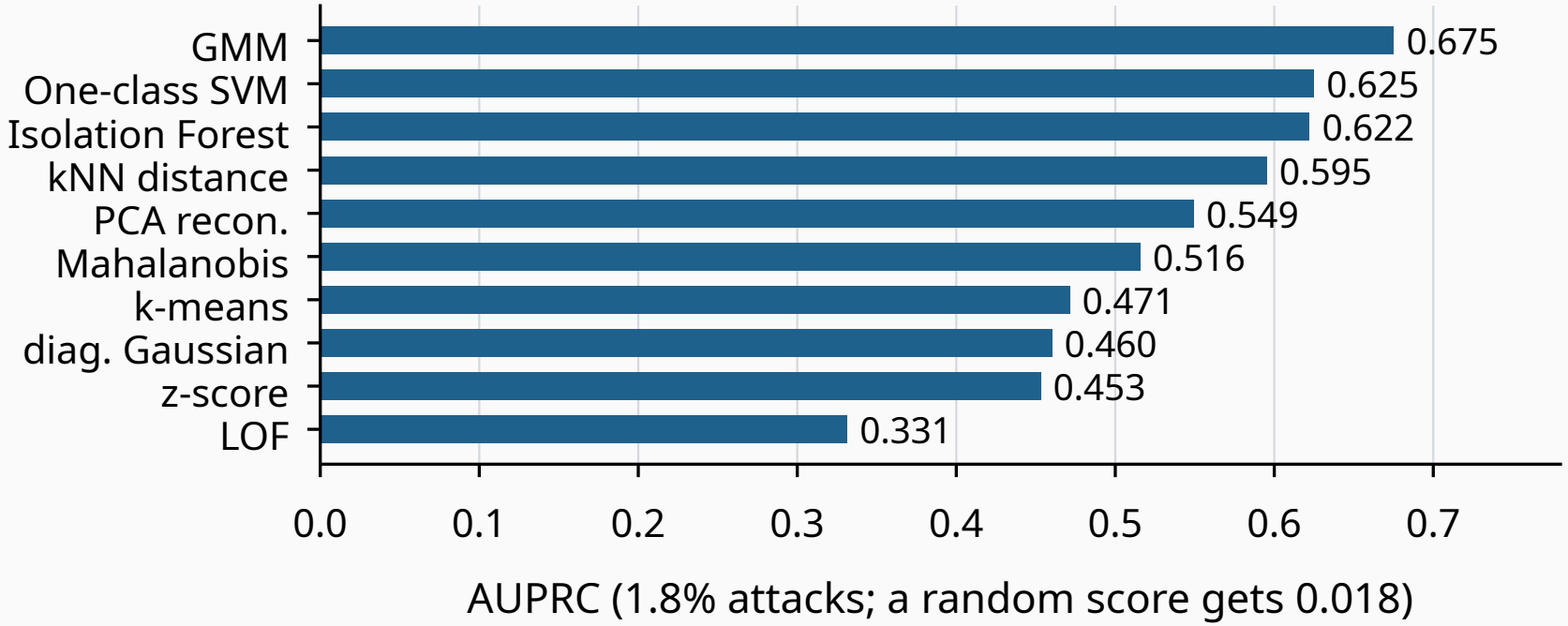

- Same scores, two test sets: 1.8% attacks, and a balanced pool.
- AUROC hardly moves; precision collapses when attacks are rare.

# DEMO 02

`notebooks/02-anomaly/notebook_02.ipynb`

# Ten detectors, one protocol

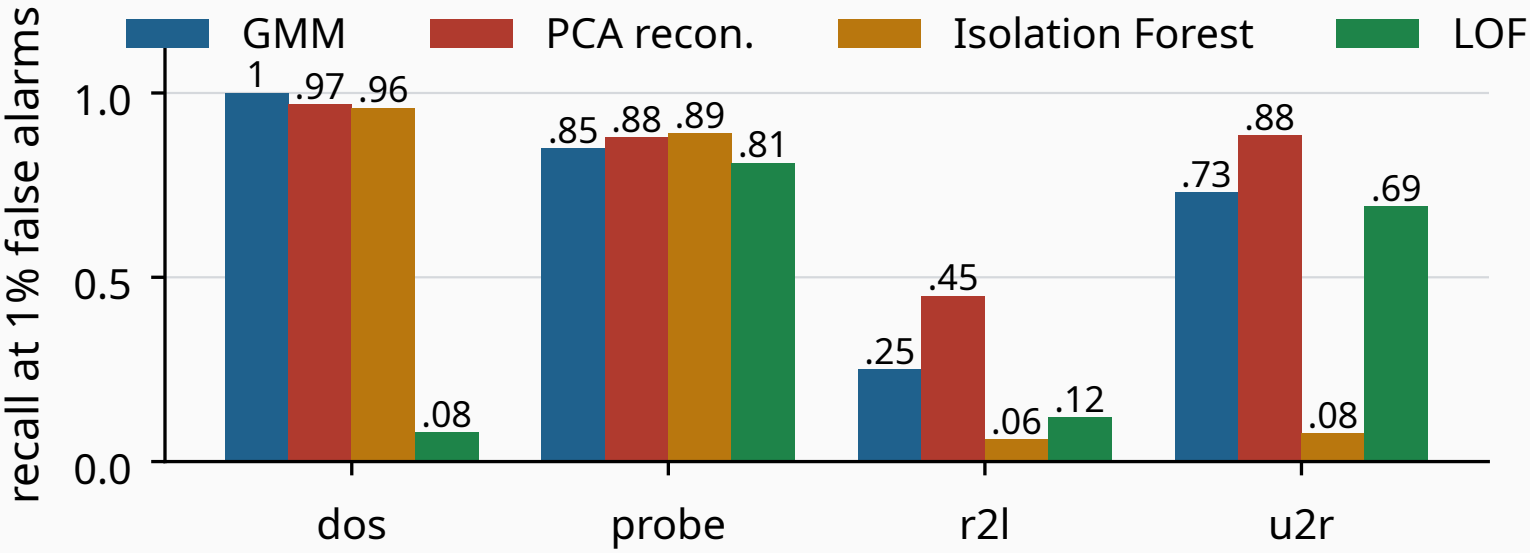
Question: which detector finds the most attacks?




- Longer = better. AUROC is 0.94-0.99 for all ten, AUPRC 0.33-0.68.
- The mixture wins; the z-score is in the lower half.

# Recall per category: one average hides the blind spots

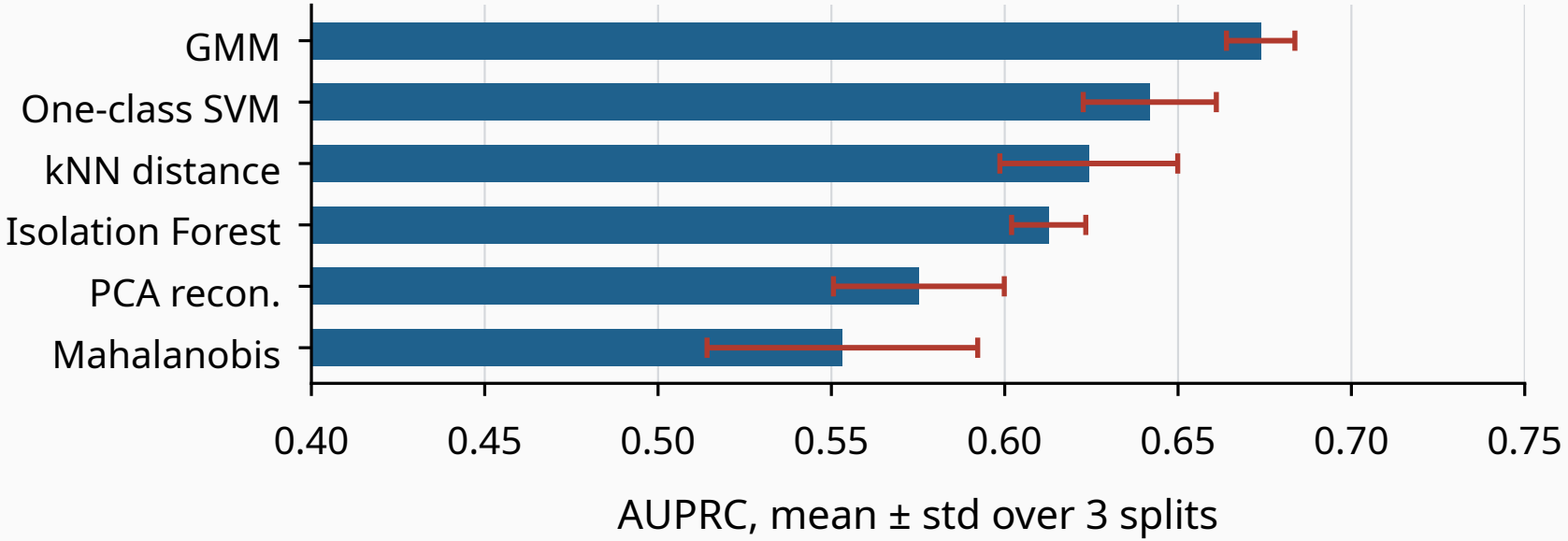
Question: which attack kinds does each detector miss?




- dos and probe are easy, except LOF on the floods (0.08).
- **r2l** (password guessing) *is* a normal-looking login: hard for all.

# Is the ranking stable? Three splits

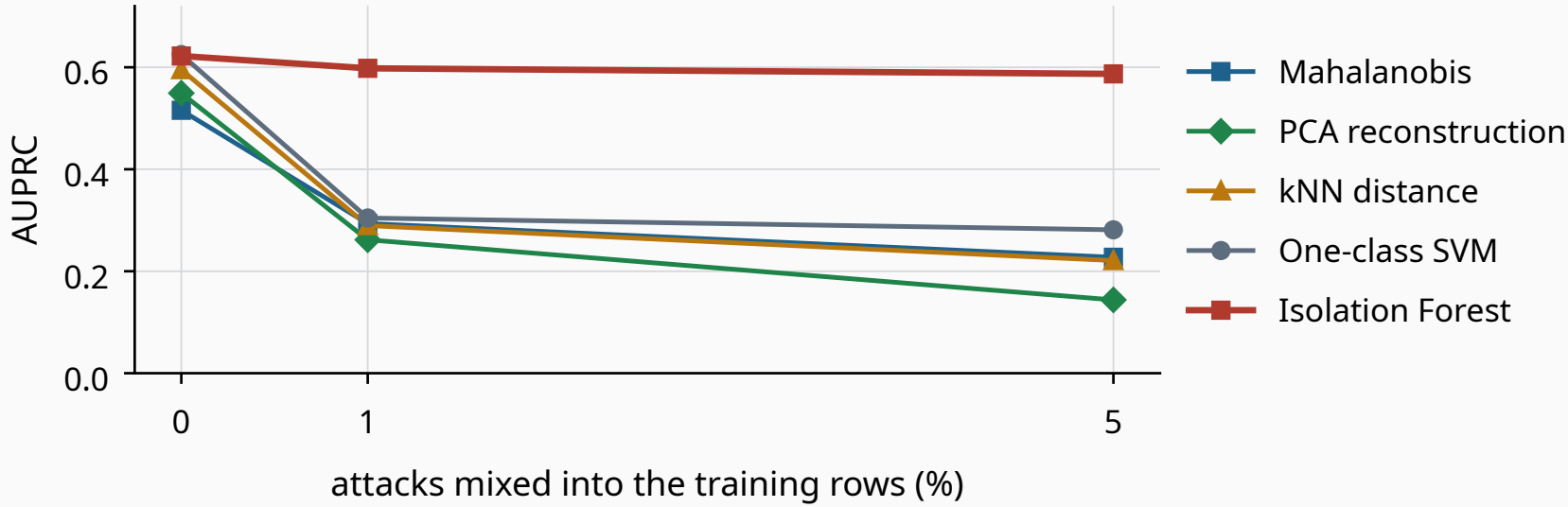
Question: does the order survive another random split?




- Mean over three splits, red whisker = one standard deviation.
- Gaps of 2-3 points are inside it: **never rank on one split** (Han et al. 2022).

# Unsupervised setting: attacks in the training data

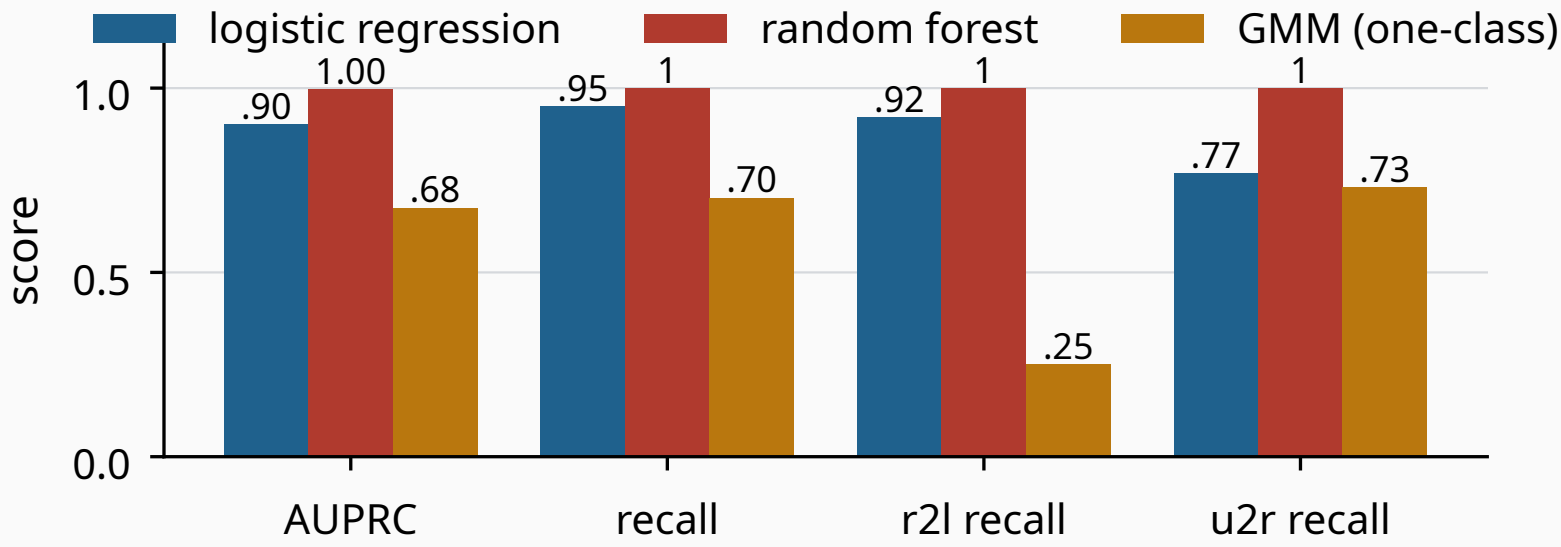
Question: what if the training rows already hold attacks?




- One line per detector, from clean rows to 1% and 5% attacks (mostly DoS).
- 1% halves most detectors; the forest barely moves (each tree sees 256 random rows).

# With labels for every attack kind, a classifier wins

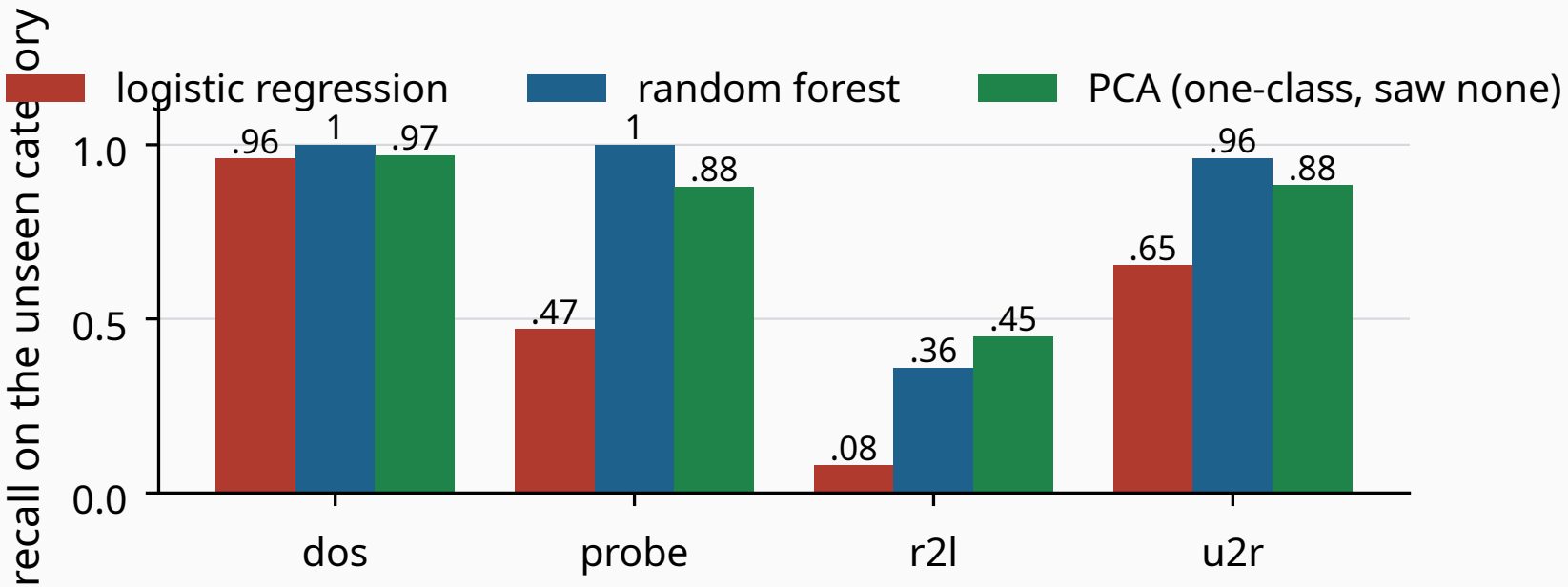
Question: how much do labelled attacks help?




- Forest AUPRC 0.998 against 0.675 for the GMM: KDD99 is “too easy” (Tavallaee et al. 2009).
- The catch: the classifier has seen *these* attacks.

# Leave one category out: the new attack

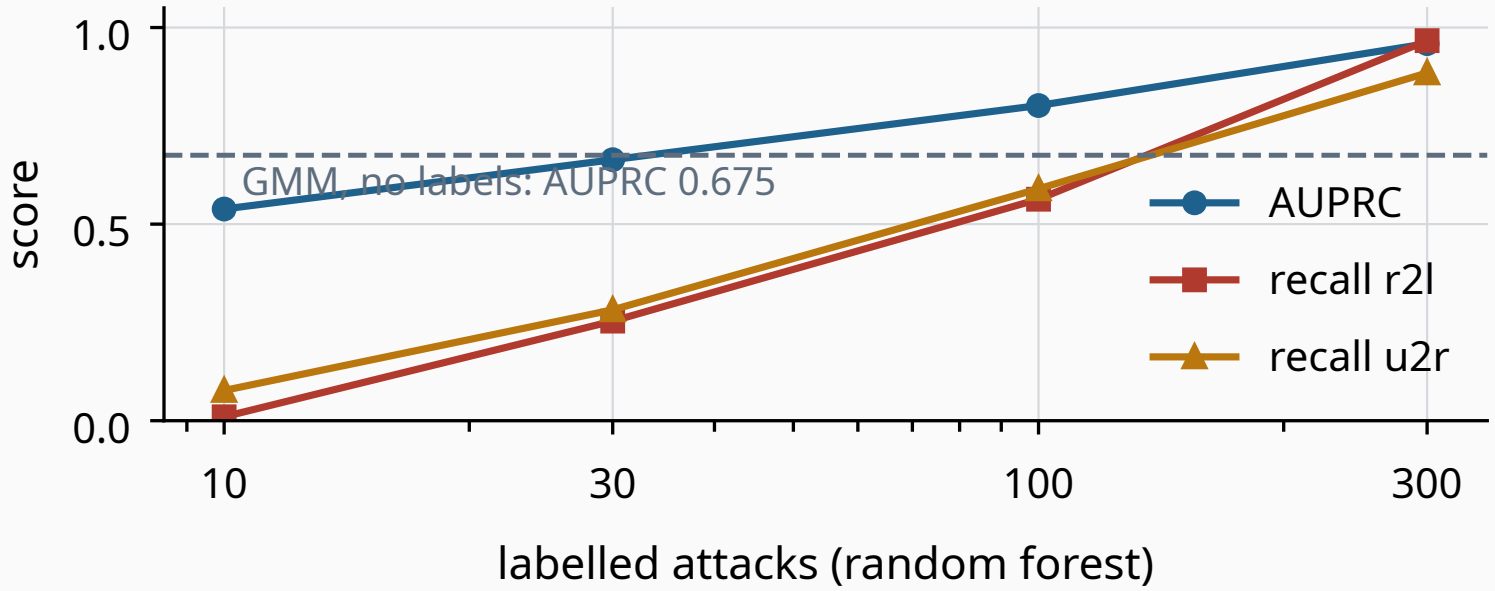
Question: how does a classifier do on an attack it never saw?




- Retrain without one category, measure recall on it.
- r2l falls on the normal side (forest 0.36); one-class PCA: 0.45.

# Few labels: the classifier needs a few hundred

Question: how many labels beat the one-class detector?




- Forest, labels from the pool (95% DoS), mean of 3: it passes the GMM at 30 to 100 labels.
- r2l and u2r need about 300 labels (Ruff et al. 2020).

# Labs and practice guide

| Notebook | Content |
|---|---|
| **00** | Example 1: z-score vs Mahalanobis, thresholds, base rate, MCD, kNN / LOF / forest |
| **01** | KDD99: duplicates, features, protocol, ROC vs precision-recall |
| **02** | ten detectors, per category, three splits, contamination, explanations |
| **03** | classifiers vs one-class: all labels, leave one category out, few labels |

- **Guide** `practice/anomaly/guide_03_01`: z-score and Mahalanobis by hand, threshold rule, AUPRC, the detector zoo, leave one category out.

# Take-aways

- Model **normal**, score the departure, set the threshold on **normal** data; the **base rate** decides what an alarm is worth.
- Six families: distribution, distance and density, isolation, boundary, reconstruction, clusters.
- Each has a blind spot (dense floods for LOF, correlations for axis cuts); none wins everywhere.
- The covariance matters: Mahalanobis sees broken relations that per-feature z-scores miss.
- Report AUPRC, per-category recall and the spread over splits; deduplicate first.
- Classifiers catch what they have seen; one-class detectors are for the rest. Use both.

# Why $d_M^2$ follows a $\chi_d^2$ distribution

- Let $\mathbf{x} \sim \mathcal{N}(\mu, C)$ and $C = LL^\top$ (Cholesky). Define $\mathbf{u} = L^{-1}(\mathbf{x} - \mu)$.
- $\mathrm{Cov}(\mathbf{u}) = L^{-1}CL^{-\top} = L^{-1}LL^\top L^{-\top} = I$: the $u_j$ are independent standard normals.
- $d_M^2 = (\mathbf{x}-\mu)^\top (LL^\top)^{-1}(\mathbf{x}-\mu) = \mathbf{u}^\top\mathbf{u} = \sum_{j=1}^{d} u_j^2$.
- A sum of $d$ squared independent standard normals is $\chi_d^2$ by definition. For $d = 2$: $P(\chi_2^2 > c) = e^{-c/2}$, so $P(d_M^2 > 20) = e^{-10} = 4.5 \times 10^{-5}$ (point B).

# LOF in full

- $k$-distance of $\mathbf{o}$: distance to its $k$-th neighbour. Reachability distance: $\mathrm{rd}_k(\mathbf{x}, \mathbf{o}) = \max\{k\text{-dist}(\mathbf{o}),\ d(\mathbf{x}, \mathbf{o})\}$ (smooths small distances).
- Local reachability density: $\mathrm{lrd}(\mathbf{x}) = \left( \frac{1}{k} \sum_{\mathbf{o} \in N_k(\mathbf{x})} \mathrm{rd}_k(\mathbf{x}, \mathbf{o}) \right)^{-1}$.
- $\mathrm{LOF}_k(\mathbf{x})$ = mean of $\mathrm{lrd}(\mathbf{o}) / \mathrm{lrd}(\mathbf{x})$ over the neighbours: a ratio, so it adapts to regions of different density (its strength).
- It returns 1 inside any dense cluster, malicious or not (its weakness).

# Isolation Forest: where $c(n)$ comes from

- An isolation tree is a random binary search tree (BST); isolating a point is an **unsuccessful search** in it.
- Its mean path length is $c(n) = 2H(n-1) - 2(n-1)/n$ with $H(i) = \sum_{m=1}^{i} 1/m \approx \ln i + 0.5772$ (Euler's constant).
- Normalising by $c(n)$ makes scores comparable across subsample sizes: $E[h] \to 0 \Rightarrow s \to 1$; $E[h] = c(n) \Rightarrow s = 0.5$; $E[h] \gg c(n) \Rightarrow s \to 0$.

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